

MUHAMMAD KALEEM

AI Engineer | Machine Learning | Generative AI

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PROFESSIONAL SUMMARY

BS Computer Science graduate and AI Engineer with hands-on experience designing Retrieval-Augmented Generation (RAG) systems, LangGraph-based AI agents, and NLP/OCR pipelines for enterprise applications. Skilled in LLM orchestration using LangChain and LangGraph, and developing end-to-end AI systems using Python and FastAPI. Experienced in Machine Learning, NLP, LLMs, AI agents, and intelligent automation, with a strong focus on building practical, scalable AI solutions. Seeking an AI Engineer role to apply practical AI engineering, problem-solving, and software development skills.

PROFESSIONAL EXPERIENCE

Data Science Intern | 10Pearls, Karachi, Pakistan

June 2026

- Built a six-phase Air Quality Index (AQI) prediction system for Karachi, covering live data ingestion (OpenWeatherMap API), historical data backfill, model development, automated retraining, and dashboarding
- Developed and benchmarked multiple regression and ensemble models (Random Forest, Ridge Regression, XGBoost) for AQI forecasting, with SHAP-based explainability to surface key pollution drivers
- Automated the end-to-end data collection and retraining pipeline using GitHub Actions, reducing manual pipeline maintenance
- Delivered an interactive Streamlit dashboard for real-time AQI monitoring and model interpretability, used internally for reporting

KEY PROJECTS

Enterprise RAG Chatbot for Islamic Banking | Personal Project

Python, LangChain, ChromaDB, FastAPI, Streamlit, PyMuPDF

- Built a Retrieval-Augmented Generation chatbot over a 23-document corpus of Meezan Bank's Annual Reports, Shariah Compliance Reports, Schedule of Charges, and product FAQs
- Engineered an ingestion pipeline using PyMuPDF and LangChain with ChromaDB vector storage for semantic document retrieval
- Delivered a natural-language Q&A interface via Streamlit for Islamic banking product and Shariah-compliance queries, directly relevant to Meezan Bank's digital banking use cases

VideoDiscovery: Multimodal Video Understanding and Recommendation System | Final Year Project

Python, YOLOv8, Whisper, CLIP, FAISS, React.js

- Developed a concurrent six-model multimodal pipeline (YOLOv8, Whisper, CLIP, SlowFast, DeepFace, Places365) for zero-shot video search and discovery
- Implemented FAISS vector indexing with Llama-3.3-70B (via Groq) for natural-language query understanding, paired with a React 19 frontend

AI SQL Database Agent | Personal Project

Python, LangChain, Groq Llama 3.3, FastAPI, React, SQLAlchemy, SQLite

- Built a natural-language-to-SQL agent using tool calling and schema inspection, enabling the LLM to interpret user queries and generate context-aware SQL against relational databases
- Implemented SQL validation and execution with Groq Llama 3.3, returning relevant records and insights through a FastAPI backend and React frontend

Voice AI Patient Registration System / Personal Project
Python, FastAPI, Vapi, LLM, Voice AI, REST APIs

- Built an AI-powered voice agent that conducts natural patient registration conversations, collects and validates patient information, and guides users through the registration workflow.
- Engineered an end-to-end voice AI pipeline integrating speech recognition, LLM-based conversational reasoning, and text-to-speech for real-time patient interaction.
- Developed a FastAPI backend for agent communication, patient data processing, API integration, and workflow orchestration.

TECHNICAL SKILLS

Languages & Frameworks: Python, SQL, Java (OOP), JavaScript, React.js, C++
AI / LLM Engineering: LangChain, LangGraph, Retrieval-Augmented Generation (RAG), Prompt Engineering, Ollama, Groq API, GPT, Claude.
Machine Learning & Data Science: Scikit-learn, Pandas, NumPy, Random Forest, XGBoost, Ridge Regression, Matplotlib, Seaborn
NLP & Computer Vision: OCR (Tesseract, PyMuPDF), Whisper, CLIP, YOLOv8, OpenCV, Image Processing, Feature Detection
Backend & Data Infrastructure: FastAPI, PostgreSQL, SQLAlchemy, SQLite, ChromaDB, FAISS

EDUCATION

Bachelor of Science in Computer Science | Sukkur IBA University *Aug 2022 – May 2026*
CGPA: 3.01/4.0 | Final Year Project: VideoDiscovery —Multimodal Video Understanding and Recommendation System

ADDITIONAL INFORMATION

- Languages: English, Urdu
- Core Strengths: Analytical Thinking, Problem Solving, Cross-Functional Communication, Team Collaboration
- Professional Interests: LLM Engineering, Retrieval-Augmented Generation, AI Engineer, Applied Machine Learning

ADDRESS

- Bhattai Colony Sector B Korangi Crossing Karachi, Sindh, Pakistan